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Abstract

1. Introduction

[Draft Note] This is a temporary introduction for organizing thoughts.

The success of large language models (LLMs) in instruction following critically depends on the quality and composition of training data. In practice, practitioners often possess small amounts of expensive, high-quality real data (e.g., 1,000 human-written demonstrations) alongside vast pools of cheap, synthetically generated data (e.g., 50,000 AI-generated examples). The central challenge is: **which subset of synthetic data should we select to maximally improve model performance under computational budget constraints?**

A key insight driving recent work is that **diversity matters**: models trained on carefully selected diverse subsets often match or exceed the performance of models trained on much larger datasets (**??**). Intuitively, redundant examples waste compute by repeatedly teaching the model what it already knows, while diverse examples expand capabilities more efficiently. However, despite widespread recognition of diversity's importance, existing approaches face a fundamental limitation—they rely on **task-agnostic heuristics** to measure diversity.

Current data selection methods face a fundamental trade-off between quality and diversity. **Influence-based methods** (**???**) excel at identifying high-utility examples by estimating each training point's contribution to evaluation objectives. However, these methods are **inherently diversity-agnostic**—they rank examples independently and greedily select top-scoring instances, often resulting in redundant selections that cover similar patterns repeatedly. To address this limitation, practitioners resort to **clustering-based strategies** (**??**): first clustering the data pool using embedding similarities, then selecting examples uniformly from each cluster. While this enforces diversity, it introduces a critical weakness: **clusters are formed using task-agnostic embeddings** (e.g., sentence representations) that may not align with what matters for the evaluation objective. Consequently, the method may select examples from semantically distinct but **task-irrelevant clusters** while neglecting examples from task-critical regions of the data space. Worse, if a cluster contains predominantly low-utility examples, uniform sampling forces the selection of poor data simply to maintain diversity.

We propose a **principled framework** that naturally unifies quality and diversity **without requiring explicit clustering**. Grounded in second-order influence functions from optimization theory, our key insight is that diversity should not be measured through task-agnostic embeddings, but rather **in terms of how examples interact when jointly affecting the evaluation objective**. When a model is trained on a data subset S, second-order Taylor expansion reveals that the change in evaluation loss naturally decomposes:

$$\Delta f \approx \underbrace{\sum_{i \in S} u_i}_{\text{Individual utility (first-order)}} - \underbrace{\sum_{i,j \in S} v_i^\top H_f v_j}_{\text{Pairwise redundancy (second-order)}} \tag{1}$$

where $u_i = g_f^\top H^{-1} \nabla \ell(z_i)$ captures each example's utility and $v_i^\top H_f v_j$ measures **curvature-aware similarity** between examples i and j under the evaluation Hessian H_f. This formulation elegantly handles both concerns: the first-order term ensures high-utility selection (naturally avoiding low-quality examples), while the second-order term penalizes redundancy

in **task-relevant parameter space**. Critically, low-utility examples in redundant regions receive compounded penalties from both terms, making their selection naturally unfavorable—**no explicit clustering required**.

This perspective offers several fundamental advantages over existing approaches. **Theoretically principled:** Our diversity metric emerges naturally from optimization analysis rather than heuristic design—it is a *consequence* of rigorously analyzing how data affects model parameters, not an ad-hoc proxy. **Task-aware:** By incorporating the evaluation Hessian H_f, we measure redundancy in the space that directly determines downstream performance, not in embedding spaces that may misalign with task requirements. **Interaction-aware:** Unlike first-order methods that sum individual contributions, we explicitly model pairwise interactions through the quadratic term, capturing diminishing returns from redundant examples. **Computationally tractable:** Despite being theoretically motivated, our method scales to LLMs via low-rank Hessian approximations, requiring only standard gradient computations.

To our knowledge, we are the first to extend second-order influence functions to explicitly incorporate evaluation objective curvature. Prior work on second-order influence (**?**) focused on improving parameter change estimation for attribution tasks, using first-order chain rule to map parameter changes to test predictions. We make a fundamental extension: by formulating second-order approximations directly in evaluation space with explicit H_f terms, we unlock a new interpretation of the second-order term as a principled diversity metric. This is more than a technical contribution—it represents a conceptual shift from viewing second-order methods as "better approximations" to viewing them as revealing task-relevant structure in data relationships.

Our contributions are:

(1) A theoretically principled diversity metric derived from second-order influence functions. Unlike embedding-based or clustering heuristics, our metric measures diversity through pairwise interactions $v_i^\top H_f v_j$ in evaluation-relevant parameter space, grounded in optimization theory.

(2) The first extension of second-order influence to evaluation objectives. Prior work (**?**) applied second-order analysis to parameter estimation; we extend this to evaluation space by explicitly incorporating H_f, enabling a new interpretation of second-order terms as measuring task-specific data redundancy.

(3) Scalable implementation for LLM-scale models. Through low-rank Hessian approximations and efficient gradient projection, our method handles pools of 50K+ examples, requiring only $O(k^2)$ operations where $k \ll n$ is the projection dimension, making curvature-aware selection practical for modern LLMs.

(4) Empirical validation demonstrating the effectiveness of curvature-aware diversity for instruction tuning data selection, with ablations confirming that our method selects genuinely diverse, complementary examples.

While we focus on instruction tuning, our framework applies broadly: any setting requiring diverse subset selection to optimize a differentiable objective can benefit from curvature-aware diversity. We hope this work encourages the community to move beyond task-agnostic embedding heuristics toward principled, optimization-grounded notions of diversity that explicitly account for how examples interact in affecting model behavior.

2. Second-order Influence

Setup. Let the (average) training objective be

$$\mathcal{L}(\theta) \;=\; \frac{1}{N}\sum_{z\in\mathcal{D}_{\text{train}}} \ell(z,\theta), \qquad \mathcal{L}_S(\theta) \;=\; \frac{1}{N}\sum_{z\in S} \ell(z,\theta),$$

and define the perturbed objective (upweighting the subset S by ε)

$$\mathcal{L}_\varepsilon(\theta) \;=\; \mathcal{L}(\theta) + \varepsilon\,\mathcal{L}_S(\theta).$$

Let $\theta^\star = \arg\min_\theta \mathcal{L}(\theta)$ and $\theta_\varepsilon = \arg\min_\theta \mathcal{L}_\varepsilon(\theta)$. Define

$$g_f := \nabla_\theta f(\theta^\star), \quad H_f := \nabla_\theta^2 f(\theta^\star),$$

$$g_S := \nabla_\theta \mathcal{L}_S(\theta^\star) = \frac{1}{N}\sum_{z\in S}\nabla_\theta \ell(z,\theta^\star), \quad H_S := \nabla_\theta^2 \mathcal{L}_S(\theta^\star) = \frac{1}{N}\sum_{z\in S}\nabla_\theta^2 \ell(z,\theta^\star),$$

$$H := \nabla_\theta^2 \mathcal{L}(\theta^\star) = \frac{1}{N} \sum_{z \in \mathcal{D}_{\text{train}}} \nabla_\theta^2 \ell(z, \theta^\star).$$

Step 1: expansion of the stationarity condition. The optimality condition for θ_ε is

$$\nabla_\theta \mathcal{L}_\varepsilon(\theta_\varepsilon) = \nabla_\theta \mathcal{L}(\theta_\varepsilon) + \varepsilon \, \nabla_\theta \mathcal{L}_S(\theta_\varepsilon) = 0.$$

Write $\Delta := \theta_\varepsilon - \theta^\star$ (so $\Delta = O(\varepsilon)$), and use

$$\nabla_\theta \mathcal{L}(\theta^\star + \Delta) \approx \nabla_\theta \mathcal{L}(\theta^\star) + H\Delta = H\Delta \quad (\text{since } \nabla_\theta \mathcal{L}(\theta^\star) = 0),$$

$$\nabla_\theta \mathcal{L}_S(\theta^\star + \Delta) \approx g_S + H_S \Delta.$$

Keeping terms up to order $O(\varepsilon^2)$ yields

$$0 \approx H\Delta + \varepsilon(g_S + H_S\Delta) \quad \Longrightarrow \quad (H + \varepsilon H_S)\Delta \approx -\varepsilon g_S.$$

Using the Neumann expansion

$$(H + \varepsilon H_S)^{-1} \approx H^{-1} - \varepsilon H^{-1} H_S H^{-1},$$

we obtain

$$\Delta \approx -(H^{-1} - \varepsilon H^{-1} H_S H^{-1}) \, \varepsilon g_S = -\varepsilon H^{-1} g_S + \varepsilon^2 H^{-1} H_S H^{-1} g_S \; + \; O(\varepsilon^3).$$

Step 2: second-order Taylor expansion of $f(\theta_\varepsilon)$. A second-order expansion around $\theta^\star$ gives

$$f(\theta_\varepsilon) = f(\theta^\star + \Delta) \approx f(\theta^\star) + g_f^\top \Delta + \frac{1}{2} \Delta^\top H_f \Delta.$$

Substituting $\Delta = -\varepsilon H^{-1} g_S + \varepsilon^2 H^{-1} H_S H^{-1} g_S + O(\varepsilon^3)$ and keeping terms up to $O(\varepsilon^2)$:

$$g_f^\top \Delta = -\varepsilon \, g_f^\top H^{-1} g_S + \varepsilon^2 \, g_f^\top H^{-1} H_S H^{-1} g_S,$$

$$\frac{1}{2} \Delta^\top H_f \Delta = \frac{\varepsilon^2}{2} \, (H^{-1} g_S)^\top H_f (H^{-1} g_S).$$

Therefore,

$$f(\theta_\varepsilon) \approx f(\theta^\star) - \varepsilon \, g_f^\top H^{-1} g_S + \frac{\varepsilon^2}{2} \Big((H^{-1} g_S)^\top H_f (H^{-1} g_S) + 2 (H^{-1} g_f)^\top H_S (H^{-1} g_S) \Big),$$

where we used $g_f^\top H^{-1} H_S H^{-1} g_S = (H^{-1} g_f)^\top H_S (H^{-1} g_S)$ (symmetry).

Synergy term as pairwise interactions. Define $v_i := H^{-1} \nabla_\theta \ell(z_i, \theta^\star)$. Then

$$H^{-1} g_S = \frac{1}{N} \sum_{z_i \in S} H^{-1} \nabla_\theta \ell(z_i, \theta^\star) = \frac{1}{N} \sum_{z_i \in S} v_i.$$

Hence,

$$(H^{-1} g_S)^\top H_f (H^{-1} g_S) = \Big(\frac{1}{N} \sum_{i \in S} v_i \Big)^\top H_f \Big(\frac{1}{N} \sum_{j \in S} v_j \Big) = \frac{1}{N^2} \sum_{i \in S} \sum_{j \in S} v_i^\top H_f v_j.$$

This shows that synergy is governed by pairwise similarities $v_i^\top H_f v_j$.

Cross-dataset synergy (two groups). Let $S = S_D \cup S_H$ and $g_S = g_D + g_H$. Then

$$(H^{-1} g_S)^\top H_f (H^{-1} g_S) = (H^{-1} g_D)^\top H_f (H^{-1} g_D) + (H^{-1} g_H)^\top H_f (H^{-1} g_H) + 2 (H^{-1} g_D)^\top H_f (H^{-1} g_H).$$

Moreover, with $v_i := H^{-1} \nabla_\theta \ell(z_i, \theta^\star)$,

$$2 (H^{-1} g_D)^\top H_f (H^{-1} g_H) = \frac{2}{N^2} \sum_{i \in S_D} \sum_{j \in S_H} v_i^\top H_f v_j.$$

Low-dimensional projection approximation. Let $P \in \mathbb{R}^{k\times n}$ be a projection (with $k \ll n$). A common approximation for first-order influence is

$$g_{\text{te}}^\top H^{-1} g_{\text{tr}} \;\approx\; (Pg_{\text{te}})^\top (PHP^\top)^{-1}(Pg_{\text{tr}}).$$

For the synergy term, define the projected solve

$$u_S := (PHP^\top)^{-1}(Pg_S) \in \mathbb{R}^k,$$

so that $H^{-1}g_S \approx P^\top u_S$ in the projected subspace, and approximate

$$(H^{-1}g_S)^\top H_f (H^{-1}g_S) \;\approx\; u_S^\top (PH_fP^\top) u_S = \big((PHP^\top)^{-1}Pg_S\big)^\top (PH_fP^\top)\big((PHP^\top)^{-1}Pg_S\big).$$

3. Selecting Synthetic Data that Complements Real Data via Second-Order Influence

3.1. Motivation

Large language models are increasingly tuned using a mixture of (i) small, high-quality real data (human-written demonstrations or preferences) and (ii) large pools of synthetic data (distilled or self-generated instructions, dialogues, preference pairs). A key challenge is that *the value of a synthetic example depends on what real data is already present*. Synthetic data can be:

- **redundant**: it pushes the parameters in directions already covered by real data, yielding diminishing returns;
- **complementary**: it fills gaps left by real data and improves target evaluation performance;
- **interfering**: it interacts with real data in a way that harms the target behavior (negative transfer).

First-order influence scores are linear in the chosen subset and thus cannot represent interaction effects between real and synthetic groups. Second-order influence introduces a quadratic term that captures these interactions, providing a principled signal for *complementary synthetic subset selection*.

3.2. Training objective and data groups

Let $z = (x, y)$ be a supervised instruction-following example. Let $\mathcal{D}_{\text{real}}$ denote the real dataset and $\mathcal{P}_{\text{syn}}$ a large pool of synthetic candidates. We aim to select a subset $S \subseteq \mathcal{P}_{\text{syn}}$ under a budget constraint (e.g., $|S| \le B$ or token budget $\text{Tok}(S) \le T$).

Define the (average) training losses

$$\mathcal{L}_{\text{real}}(\theta) := \frac{1}{|\mathcal{D}_{\text{real}}|} \sum_{z\in\mathcal{D}_{\text{real}}} \ell(z,\theta), \qquad \mathcal{L}_S(\theta) := \frac{1}{|S|}\sum_{z\in S} \ell(z,\theta),$$

where for SFT we can use the standard token-level negative log-likelihood

$$\ell((x,y),\theta) := -\log p_\theta(y \,|\, x).$$

We consider a small upweighting of the synthetic subset S:

$$\mathcal{L}_\varepsilon(\theta) := \mathcal{L}_0(\theta) + \varepsilon\, \mathcal{L}_S(\theta),$$

where $\mathcal{L}_0$ is a *reference* training objective defining the reference point $\theta^\star$. Pragmatically, $\theta^\star$ can be (i) a checkpoint before adding the new synthetic data, or (ii) a solution trained on a broader mixture so that group-wise gradients need not vanish exactly. This avoids degeneracy where $\nabla\mathcal{L}_{\text{real}}(\theta^\star) = 0$ would remove real–synthetic interaction signals.

Let

$$\theta^\star := \arg\min_\theta \mathcal{L}_0(\theta), \qquad \theta_\varepsilon := \arg\min_\theta \mathcal{L}_\varepsilon(\theta).$$

3.3. Evaluation function f for LLM tuning

We choose a differentiable evaluation objective $f(\theta)$ to *minimize*, defined on held-out evaluation sets. A practical choice is a weighted sum of differentiable surrogate losses:

$$f(\theta) = \lambda_{\mathrm{ID}} f_{\mathrm{ID}}(\theta) + \lambda_{\mathrm{OOD}} f_{\mathrm{OOD}}(\theta) + \lambda_{\mathrm{safe}} f_{\mathrm{safe}}(\theta) + \lambda_{\mathrm{pref}} f_{\mathrm{pref}}(\theta),$$

where:

$$f_{\mathrm{ID}}(\theta) := \frac{1}{|\mathcal{D}_{\mathrm{eval}}^{\mathrm{ID}}|} \sum_{(x,y)\in\mathcal{D}_{\mathrm{eval}}^{\mathrm{ID}}} \big(-\log p_\theta(y|x)\big),$$

$$f_{\mathrm{OOD}}(\theta) := \frac{1}{|\mathcal{D}_{\mathrm{eval}}^{\mathrm{OOD}}|} \sum_{(x,y)\in\mathcal{D}_{\mathrm{eval}}^{\mathrm{OOD}}} \big(-\log p_\theta(y|x)\big),$$

$$f_{\mathrm{safe}}(\theta) := \frac{1}{|\mathcal{D}_{\mathrm{eval}}^{\mathrm{safe}}|} \sum_{(x,y_{\mathrm{safe}})\in\mathcal{D}_{\mathrm{eval}}^{\mathrm{safe}}} \big(-\log p_\theta(y_{\mathrm{safe}}|x)\big).$$

If preference data is available, a differentiable held-out preference objective (e.g., DPO-style) can be used:

$$f_{\mathrm{pref}}(\theta) := \frac{1}{|\mathcal{D}_{\mathrm{eval}}^{\mathrm{pref}}|} \sum_{(x,y^+,y^-)\in\mathcal{D}_{\mathrm{eval}}^{\mathrm{pref}}} \Big(-\log \sigma\big(\beta(\log p_\theta(y^+|x) - \log p_\theta(y^-|x))\big)\Big),$$

with inverse-temperature $\beta > 0$ and sigmoid $\sigma(\cdot)$. The weights $\lambda_\cdot$ set the trade-off between in-domain quality, generalization, safety, and preferences.

3.4. Second-order influence for adding a synthetic subset

Define at $\theta^\star$:

$$g_f := \nabla_\theta f(\theta^\star), \quad H_f := \nabla_\theta^2 f(\theta^\star), \quad H := \nabla_\theta^2 \mathcal{L}_0(\theta^\star),$$

and for the selected synthetic subset S,

$$g_S := \nabla_\theta \mathcal{L}_S(\theta^\star), \quad H_S := \nabla_\theta^2 \mathcal{L}_S(\theta^\star).$$

The second-order influence expansion gives

$$f(\theta_\varepsilon) \approx f(\theta^\star) - \varepsilon\, g_f^\top H^{-1} g_S + \frac{\varepsilon^2}{2}\Big((H^{-1}g_S)^\top H_f (H^{-1}g_S) + 2(H^{-1}g_f)^\top H_S (H^{-1}g_S)\Big). \tag{2}$$

In what follows, we focus on the synergy term

$$\Psi(S) := (H^{-1}g_S)^\top H_f (H^{-1}g_S),$$

which captures subset interactions and admits a pairwise expansion.

3.5. Pairwise interaction view and complementarity

Let $v(z) := H^{-1}\nabla_\theta \ell(z,\theta^\star)$. For $S = \{z_1,\dots,z_m\}$ and $g_S = \frac{1}{m}\sum_{i=1}^m \nabla_\theta \ell(z_i,\theta^\star)$,

$$H^{-1}g_S = \frac{1}{m}\sum_{i=1}^m v(z_i), \qquad \Psi(S) = \Big(\frac{1}{m}\sum_{i=1}^m v(z_i)\Big)^\top H_f \Big(\frac{1}{m}\sum_{j=1}^m v(z_j)\Big) = \frac{1}{m^2}\sum_{i,j=1}^m v(z_i)^\top H_f v(z_j).$$

Thus, the synergy is governed by pairwise similarities under the curvature induced by H_f.

Real–synthetic complementarity via a cross term. Let R denote a fixed real anchor group (e.g., $\mathcal{D}_{\text{real}}$ or a real minibatch distribution), with

$$g_R := \nabla_\theta \mathcal{L}_R(\theta^\star), \qquad \text{and} \qquad g_{R\cup S} = g_R + g_S.$$

Then the synergy decomposes as

$$\Psi(R\cup S) = (H^{-1}g_R)^\top H_f (H^{-1}g_R) + (H^{-1}g_S)^\top H_f (H^{-1}g_S) + 2(H^{-1}g_R)^\top H_f (H^{-1}g_S). \tag{3}$$

The cross term

$$\Psi_\times(R,S) := (H^{-1}g_R)^\top H_f (H^{-1}g_S)$$

quantifies how synthetic updates interact with real updates under the evaluation curvature.

Sign convention (minimization). Because f is defined as an evaluation *loss* to be minimized, the net predicted gain from adding S includes a term $-\varepsilon^2\,\Psi_\times(R,S)$ (see below). Hence, for loss-minimization, a *negative* $\Psi_\times(R,S)$ is constructive (complementary), while a positive $\Psi_\times(R,S)$ is potentially interfering.

3.6. Objective for selecting complementary synthetic subsets

We compare training with real anchor R alone versus $R\cup S$. Ignoring the H_S-dependent correction in (2) for clarity (or treating it as a higher-cost optional term), the predicted change in evaluation loss from adding S on top of R is

$$\begin{aligned}\widehat{\Delta f}(S;R) &:= \widehat{f}(R\cup S) - \widehat{f}(R)\\ &\approx -\varepsilon\, g_f^\top H^{-1} g_S + \frac{\varepsilon^2}{2}\Big(2(H^{-1}g_R)^\top H_f (H^{-1}g_S) + (H^{-1}g_S)^\top H_f (H^{-1}g_S)\Big).\end{aligned} \tag{4}$$

Define the predicted *gain* (decrease in f)

$$\widehat{\mathcal{G}}(S;R) := -\widehat{\Delta f}(S;R).$$

Then

$$\widehat{\mathcal{G}}(S;R) \approx \underbrace{\varepsilon\, g_f^\top H^{-1} g_S}_{\text{first-order utility}} - \underbrace{\varepsilon^2 (H^{-1}g_R)^\top H_f (H^{-1}g_S)}_{\text{real–synthetic complementarity}} - \underbrace{\frac{\varepsilon^2}{2}(H^{-1}g_S)^\top H_f (H^{-1}g_S)}_{\text{synthetic redundancy / diminishing returns}}. \tag{5}$$

We select S by maximizing $\widehat{\mathcal{G}}(S;R)$ subject to a budget:

$$\max_{S\subseteq \mathcal{P}_{\text{syn}}} \widehat{\mathcal{G}}(S;R) \quad \text{s.t.} \quad |S| \le B \;\text{ or }\; \text{Tok}(S) \le T.$$

Per-example decomposition. Let $S = \{z_{i_1}, \dots, z_{i_m}\}$ and define $v_i := v(z_i) = H^{-1}\nabla \ell(z_i, \theta^\star)$. Let

$$u_i := g_f^\top v_i, \qquad c_i := (H^{-1}g_R)^\top H_f v_i, \qquad q_{ij} := v_i^\top H_f v_j.$$

Then

$$\widehat{\mathcal{G}}(S;R) \approx \frac{\varepsilon}{m}\sum_{i\in S} u_i - \frac{\varepsilon^2}{m}\sum_{i\in S} c_i - \frac{\varepsilon^2}{2m^2}\sum_{i,j\in S} q_{ij}.$$

The last term penalizes choosing many synthetic points that are similar under q_{ij}, which operationalizes a curvature-aware notion of redundancy in the Hessian-weighted gradient space.

Greedy selection (practical). A practical approach is greedy maximization using marginal gains: for current set S, add z_i that maximizes $\Delta(i \mid S) := \widehat{\mathcal{G}}(S\cup\{z_i\};R) - \widehat{\mathcal{G}}(S;R)$. This yields an interaction-aware data selection rule that prefers high first-order utility, negative cross interaction with R (constructive complementarity for loss-minimization), and low redundancy with the already selected synthetic points.

3.7. Projected computation for LLM-scale models

Directly forming H^{-1} and H_f is infeasible for LLMs. Introduce a projection $P \in \mathbb{R}^{k\times n}$ with $k \ll n$ and define

$$\tilde{H} := PHP^\top, \quad \tilde{H}_f := PH_fP^\top, \quad \tilde{g}(z) := P\nabla_\theta \ell(z,\theta^\star), \quad \tilde{g}_f := Pg_f, \quad \tilde{g}_R := Pg_R.$$

Approximate

$$H^{-1}g \approx P^\top \tilde{H}^{-1}(Pg), \qquad v(z) = H^{-1}\nabla \ell(z,\theta^\star) \approx P^\top \tilde{H}^{-1}\tilde{g}(z).$$

Then the key quantities become low-dimensional:

$$u_i \approx \tilde{g}_f^\top \tilde{H}^{-1}\tilde{g}(z_i), \qquad c_i \approx (\tilde{H}^{-1}\tilde{g}_R)^\top \tilde{H}_f(\tilde{H}^{-1}\tilde{g}(z_i)), \qquad q_{ij} \approx (\tilde{H}^{-1}\tilde{g}(z_i))^\top \tilde{H}_f(\tilde{H}^{-1}\tilde{g}(z_j)).$$

All computations scale with k rather than n.

3.8. What to expect (testable hypotheses)

Assuming the approximation is sufficiently accurate for ranking, we expect:

1. **Token efficiency:** at a fixed synthetic token budget, the selected set S yields lower evaluation loss (or better downstream metrics) than baselines such as random selection and first-order influence-only selection.
2. **Complementarity over redundancy:** selected synthetic examples exhibit more negative cross interaction $\Psi_\times(R,S)$ (constructive for loss-minimization) compared to naive diversity heuristics, and lower within-synthetic redundancy measured by the quadratic term.
3. **Improved generalization:** when f includes an OOD component, selection preferentially chooses synthetic data that improves OOD while minimally degrading in-domain and safety components.
4. **Predictive validity:** the predicted gain $\widehat{\mathcal{G}}(S;R)$ correlates with empirical improvements measured after fine-tuning (rank correlation across different subsets).

3.9. Experimental plan

Step 0: model and training regime. Choose a base LLM and a tuning regime (full fine-tuning or parameter-efficient tuning such as LoRA). Fix the real dataset $\mathcal{D}_{\text{real}}$ and prepare a large synthetic pool $\mathcal{P}_{\text{syn}}$.

Step 1: define evaluation objective f. Construct evaluation sets: $\mathcal{D}_{\text{eval}}^{\text{ID}}$ (held-out real), $\mathcal{D}_{\text{eval}}^{\text{OOD}}$ (gap-focused tasks), $\mathcal{D}_{\text{eval}}^{\text{safe}}$ (safety), and optionally $\mathcal{D}_{\text{eval}}^{\text{pref}}$ (preferences). Define f as the weighted sum in the previous section.

Step 2: choose reference point $\theta^\star$. Use a checkpoint that reflects the current model state before adding the candidate synthetic subset. Compute (or approximate) H, g_f, H_f, and the group gradient g_R at $\theta^\star$.

Step 3: compute projected quantities. Choose projection dimension k and projection matrix P. Compute $\tilde{H}$ and $\tilde{H}_f$ using stochastic approximations (e.g., Gauss–Newton / Fisher-style estimates). For each candidate synthetic example $z_i \in \mathcal{P}_{\text{syn}}$, compute $\tilde{g}(z_i)$.

Step 4: selection methods (compare baselines). For each budget B (or token budget T), construct subsets:

- **Random:** uniform sampling from $\mathcal{P}_{\text{syn}}$.
- **First-order only:** select top-B by u_i (utility) or by $g_f^\top H^{-1} g_{z_i}$.
- **Heuristic diversity:** cluster-based uniform sampling (embedding-space or gradient-space).
- **Second-order complement (proposed):** greedy maximize $\widehat{\mathcal{G}}(S;R)$ in (5) using projected computations.
- **Ablations:** (i) remove cross term; (ii) remove redundancy term; (iii) vary k.

Step 5: controlled fine-tuning. For each selected subset S, fine-tune from the same initialization with the same compute budget. Keep the real data proportion fixed and add synthetic subset S according to a controlled schedule.

Step 6: evaluation and analysis. Report:

- **Primary:** evaluation losses corresponding to $f_{\mathrm{ID}}, f_{\mathrm{OOD}}, f_{\mathrm{safe}}, f_{\mathrm{pref}}$.
- **Secondary:** non-differentiable downstream metrics (benchmark scores, judge win-rates, red-teaming success rates).
- **Attribution validity:** rank correlation between $\widehat{\mathcal{G}}(S; R)$ and empirical gains.
- **Interaction diagnostics:** distributions of $\Psi_{\times}(R, S)$ and within-synthetic redundancy for each method.

Step 7: robustness checks. Repeat across: (i) different synthetic generators, (ii) different real dataset sizes (to test complementarity under scarcity), (iii) different evaluation weightings $\lambda_{\cdot}$, (iv) multiple random seeds for selection and training.

4. Influence Estimation with Non-differentiable Evaluation Metrics

4.1. Why non-differentiable f arises in LLM evaluation

Many LLM evaluations are *discrete* or *procedural*: exact match, pass@k, unit-test success, toxicity thresholding, and win-rate under an external judge (human or LLM-as-judge). Such metrics are non-differentiable with respect to model parameters θ, especially under deterministic decoding. However, the second-order influence expansion used in this work requires $g_f = \nabla_\theta f(\theta^\star)$ and $H_f = \nabla^2_\theta f(\theta^\star)$ in Eq. (1). This section provides two ways to handle non-differentiable f:

- **Stochastic relaxation (differentiable in expectation):** define f as the expected value of a non-differentiable metric under stochastic generation, then compute unbiased estimators of g_f and (if needed) H_f via score-function identities.
- **Black-box directional estimation:** avoid explicit g_f and H_f by estimating the directional derivatives of f along influence directions using a small number of function evaluations.

4.2. Stochastic relaxation: expected metric objective

Let $\mathcal{D}_{\mathrm{eval}} = \{x_i\}_{i=1}^M$ be evaluation prompts. Let the model induce a conditional distribution over outputs $y \sim p_\theta(\cdot \mid x)$ (e.g., sampling at temperature $\tau > 0$). Let $r(x, y) \in \mathbb{R}$ be a (possibly non-differentiable) reward computed from a metric or a judge, for example:

- $r(x, y) = \mathbf{1}\{\text{unit tests pass}\}$ for code,
- $r(x, y) = \mathbf{1}\{\text{exact match}\}$ for QA,
- $r(x, y) = \mathbf{1}\{\text{judge prefers } y \text{ over baseline}\}$ for win-rate,
- $r(x, y) \in \{0, 1, 2\}$ for graded judge outcomes.

Define the evaluation *loss* to minimize as the negative expected reward:

$$f(\theta) := -\frac{1}{M} \sum_{i=1}^{M} \mathbb{E}_{y \sim p_\theta(\cdot \mid x_i)} \big[r(x_i, y) \big]. \tag{6}$$

Even if $r(\cdot, \cdot)$ is non-differentiable, $f(\theta)$ is differentiable under mild regularity conditions, because θ only enters through $p_\theta(y \mid x)$.

First derivative (score-function / REINFORCE). Let $s_\theta(x, y) := \nabla_\theta \log p_\theta(y \mid x)$. Then

$$g_f(\theta) = \nabla_\theta f(\theta) = -\frac{1}{M} \sum_{i=1}^{M} \mathbb{E}_{y \sim p_\theta(\cdot \mid x_i)} \Big[r(x_i, y)\, s_\theta(x_i, y) \Big]. \tag{7}$$

A standard variance-reduced form introduces a baseline $b(x)$:

$$g_f(\theta) = -\frac{1}{M} \sum_{i=1}^{M} \mathbb{E}_{y \sim p_\theta(\cdot \mid x_i)} \Big[(r(x_i, y) - b(x_i))\, s_\theta(x_i, y) \Big],$$

since $\mathbb{E}[b(x)\, s_\theta(x, y)] = b(x) \nabla_\theta \sum_y p_\theta(y \mid x) = 0$.

Second derivative (Hessian identity). Assuming $r(x, y)$ does not depend on θ (judge fixed),

$$\begin{aligned} H_f(\theta) &= \nabla_\theta^2 f(\theta) \\ &= -\frac{1}{M} \sum_{i=1}^{M} \mathbb{E}_{y \sim p_\theta(\cdot \mid x_i)} \Big[r(x_i, y)\, \nabla_\theta^2 \log p_\theta(y \mid x_i) + r(x_i, y)\, s_\theta(x_i, y) s_\theta(x_i, y)^\top \Big]. \end{aligned} \tag{8}$$

In practice, one rarely forms H_f explicitly; one estimates Hessian-vector products or only the quadratic forms $u^\top H_f u$ needed by second-order influence.

4.3. Plug-in influence with expected metric f

Let $\theta^\star$ be the reference parameters and let $H = \nabla_\theta^2 \mathcal{L}_0(\theta^\star)$. For a training subset S with $g_S = \nabla_\theta \mathcal{L}_S(\theta^\star)$ and $H_S = \nabla_\theta^2 \mathcal{L}_S(\theta^\star)$, the second-order influence approximation (Eq. (1)) applies with f defined by (6):

$$f(\theta_\varepsilon) \approx f(\theta^\star) - \varepsilon\, g_f^\top H^{-1} g_S + \frac{\varepsilon^2}{2} \Big((H^{-1} g_S)^\top H_f (H^{-1} g_S) + 2 (H^{-1} g_f)^\top H_S (H^{-1} g_S) \Big), \tag{9}$$

where g_f and H_f can be estimated via (7)–(8) at $\theta = \theta^\star$.

Quadratic form of H_f without constructing H_f. Let $u \in \mathbb{R}^n$. From (8),

$$u^\top H_f(\theta) u = -\frac{1}{M} \sum_{i=1}^{M} \mathbb{E}_{y \sim p_\theta(\cdot \mid x_i)} \Big[r(x_i, y)\, u^\top \nabla_\theta^2 \log p_\theta(y \mid x_i)\, u + r(x_i, y)\, (u^\top s_\theta(x_i, y))^2 \Big]. \tag{10}$$

Thus the synergy term in (9) can be computed using only: (i) scalar products $u^\top s_\theta$, and (ii) Hessian-vector products of $\nabla_\theta^2 \log p_\theta(y \mid x)$ with u.

4.4. Black-box influence estimation via directional finite differences

If the evaluation metric is available only as a black-box function (e.g., proprietary judge, human-in-the-loop, complex simulator), we can avoid explicit g_f and H_f by estimating the required directional derivatives.

Influence direction. Define the (first-order) influence direction for subset S:

$$d_S := -H^{-1} g_S. \tag{11}$$

This is the parameter change induced by upweighting S by ε to first order. (For LLMs, d_S can be approximated in a low-dimensional subspace via gradient projection.)

First-order influence as a directional derivative. Assuming f is directionally differentiable at $\theta^\star$ along d_S,

$$-\varepsilon\, g_f^\top H^{-1} g_S = \varepsilon\, g_f^\top d_S = \varepsilon \frac{\mathrm{d}}{\mathrm{d}t} f(\theta^\star + t d_S) \Big|_{t=0}. \tag{12}$$

We can estimate the derivative using a one-sided finite difference:

$$\widehat{\mathcal{I}}^{(1)}(S) := \varepsilon \cdot \frac{f(\theta^\star + h d_S) - f(\theta^\star)}{h}, \tag{13}$$

with step size $h > 0$.

Second-order synergy as a second directional derivative. The synergy term equals the second directional derivative of f along d_S:

$$(H^{-1}g_S)^\top H_f(H^{-1}g_S) = d_S^\top H_f d_S = \frac{\mathrm{d}^2}{\mathrm{d}t^2} f(\theta^\star + t d_S)\Big|_{t=0}. \tag{14}$$

A centered finite difference yields:

$$\widehat{\Psi}(S) := \frac{f(\theta^\star + h d_S) - 2f(\theta^\star) + f(\theta^\star - h d_S)}{h^2}. \tag{15}$$

Then the second-order contribution in Eq. (1) can be approximated as $\frac{\varepsilon^2}{2}\widehat{\Psi}(S)$ (ignoring the H_S correction term), or used directly as an interaction-aware selection score.

Cross interaction between real anchor R and synthetic subset S. For complementarity analysis, the cross term is the mixed second derivative:

$$(H^{-1}g_R)^\top H_f(H^{-1}g_S) = d_R^\top H_f d_S = \frac{\partial^2}{\partial\alpha\,\partial\beta} f(\theta^\star + \alpha d_R + \beta d_S)\Big|_{\alpha=\beta=0}. \tag{16}$$

It can be estimated via a four-point stencil:

$$\widehat{\Psi}_\times(R,S) := \frac{f(\theta^\star + h d_R + h d_S) - f(\theta^\star + h d_R) - f(\theta^\star + h d_S) + f(\theta^\star)}{h^2}. \tag{17}$$

This estimator uses only evaluations of f, not its gradients.

4.5. Practical notes (variance and feasibility)

Common random numbers. When $f(\theta)$ is estimated by sampling generations and judging them, use the same random seeds (or the same sampled noise) across $\theta^\star$, $\theta^\star \pm h d_S$, and $\theta^\star + h(d_R + d_S)$. This coupling greatly reduces the variance of finite differences.

Where to perturb parameters. For large LLMs, apply perturbations only in a tractable subspace: adapter weights (LoRA) or a projected subspace $P^\top\mathbb{R}^k$. Then d_S in (11) is replaced by a projected direction, and $\theta^\star \pm h d_S$ can be realized by a low-rank parameter delta.

Step-size selection. Pick h such that: (i) the metric estimator is stable (not dominated by Monte Carlo noise), and (ii) the Taylor approximation remains locally valid. A practical procedure is to sweep h over a logarithmic grid and choose a region where the finite-difference estimate is stable.

Summary. Non-differentiable evaluations can be handled either by: (i) defining an expected-metric f and using score-function estimators for g_f, H_f, or (ii) treating f as black-box and estimating the first-order and second-order influence terms from a small number of metric evaluations along influence directions.